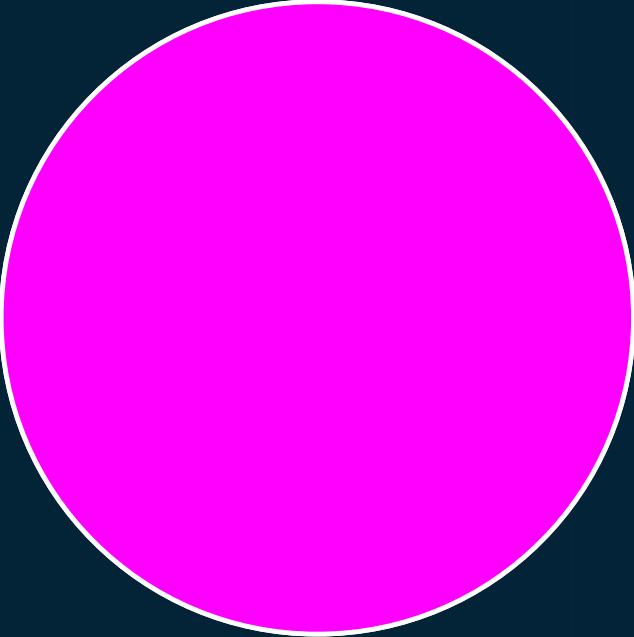




AI Lifecycle Security & Human Oversight

Chapter 05 · Episode 01 · Domain 1.3 — The MDLC End-to-End



Learning Objectives

1. Map all seven MDLC phases to their primary controls
2. Distinguish HITL, HOTL, and governance oversight by risk
3. Apply lifecycle risk mapping to exam scenarios

Phases 1–2: Use Case & Data

PHASE 1 · USE CASE

Risk-appetite definition

Impact assessment (required)

Regulatory scoping

No documented purpose = unmanaged risk

PHASE 2 · DATA

Classification & minimization

Integrity & verification

Provenance

Model security is decided here

Phases 3–4: Development & TEVV

PHASE 3 · DEVELOPMENT

Threat-model the architecture
OWASP LLM Top 10 · MITRE ATLAS
Secure training infra
Version control + checkpoint signing

PHASE 4 · TEVV ★

Accuracy + bias testing
Adversarial robustness
Hallucination testing
Fail TEVV → no deployment

Phases 5–7: Deploy, Monitor, Iterate

PHASE 5 · DEPLOYMENT

RBAC + encryption

Prompt guardrails

Canary rollout + auto-rollback

The attack surface is now live

PHASES 6–7 · RUN

Drift & log monitoring

Cost/abuse signals + IR

Human review loops

Retrain → re-run TEVV

Three Human-Oversight Models



HITL

HIGH risk

Human approves before action

Identity, finance, discipline



HOTL

MEDIUM risk

AI acts, human can intervene

Fraud, alert triage

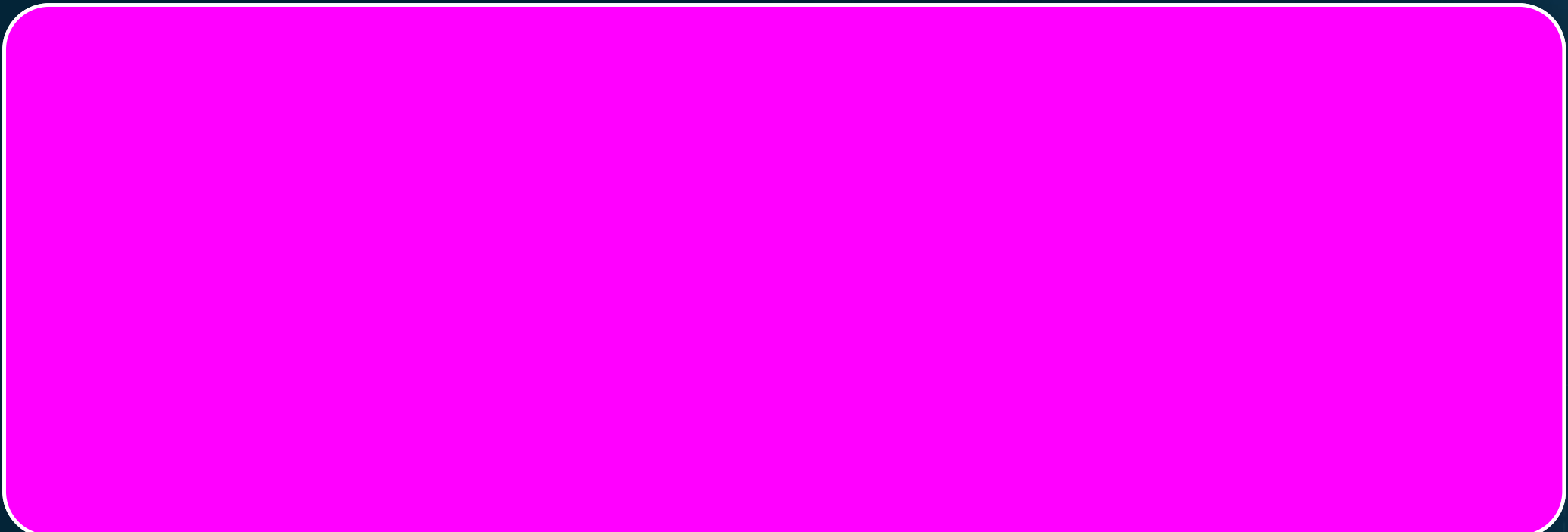


GOVERNANCE

LOW risk

Humans set policy only

Log/spam classification



Demo: Lifecycle Risk Scanner

Scans a model config against all seven MDLC phases — a Phase-4 TEVV failure blocks the pipeline

Lifecycle Risk Mapping (Scenario → Phase)



BLIND SPOTS

→ Phase 2 data balancing



UNDISCOVERED BIAS

→ Phase 4 TEVV



DRIFT AFTER CHANGE

→ Phase 6 monitoring



SHADOW DEPLOY

→ Phase 1 use case

Exam Tips



MDLC

Organizes all of Domain 1



OVERSIGHT

HITL/HOTL/gov by risk



TEVV

A gate, not a checkbox



SECURITY

At every phase